

Final Project Report: Functional Calculator

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Project: Functional Calculator

1. Introduction

The main goal of this project was to build a functional calculator that uses Python's functional programming concepts. Instead of just doing basic math , I wanted to see how we could use Lambda expressions , Dictionaries, and higher order functions like map and filter to make the code cleaner, shorter, and more efficient.

2. Challenges and Solutions

- **Division by Zero:** When I first tested the calculator , it would crash if the user tried to divide by zero. I fixed this by adding a simple if-else statement inside the Lambda expression to return an error message instead of letting the program break.
- **History Display:** I needed a way to track and display the user's past calculations. I used a list to store the strings of the equations and the enumerate() function to display them with proper numbering. This was my first time using enumerate , but it turned out to be really straightforward.
- **Lambda & Dictionaries:** At first, it was a bit confusing to link Lambda functions inside a dictionary. After reading a bit more about it, I realized it's actually a great way to avoid writing long chains of if-elif statements.

3. Required Concepts Used

The project covers almost everything we learned this semester:

- **Data Types:** Using floats, strings, and f-strings for formatting.
- **Control Flow:** Used while loops to keep the program running and if-else for the menu and logic.
- **Data Structures:** Used lists for history, sets for unique results , and dictionaries to map operators.
- **Functional Concepts:** Used map() to double list items, filter() to get positive numbers , zip() to add lists together, and enumerate() for history indexing.

4. Conclusion

Building this calculator helped me understand how Python handles functional programming. It showed me that coding isn't just about making something that works it's about writing "clean" code that is easy to read and maintain.